Quiz 1

Reg no: **Section:** **Date:**

Q1 . Suppose you have a transportation network with 10 cities represented by the adjacency matrix below. Each connection between cities has a specific weight (representing distance in kilometers). The goal is to analyze paths between City A and City J using different search algorithms.

	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**
A	0	2	4	0	0	0	0	0	0	0
B	2	0	0	7	0	3	0	0	0	0
C	4	0	0	1	5	0	0	0	0	0
D	0	7	1	0	0	0	2	6	0	0
E	0	0	5	0	0	0	0	0	4	0
F	0	3	0	0	0	0	8	0	0	5
G	0	0	0	2	0	8	0	0	0	3
H	0	0	0	6	0	0	0	0	1	0
I	0	0	0	0	4	0	0	1	0	2
J	0	0	0	0	0	5	3	0	2	0

1. Using BFS, what is the shortest path from City A to City J (considering unweighted paths)?
2. Using DFS, find the longest path from City A to City J without revisiting any city.
3. Using the Greedy Algorithm (always choose the nearest next city), what path would you find from City A to City J? Is this the optimal path? Justify your answer.

Q.2 Solve the following by using the Hill climbing algorithm.

2	3	
1	8	4
7	6	5

Initial state

1	2	3
8		4
7	6	5

Goal state